

## COURSE SYLLABUS

# Querying and Analyzing SQL Server 2025

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COURSE NUMBER

**65317A**

DURATION

**3 days · 10 modules**

AUDIENCE LEVEL

**Beginner**

FORMAT

**Instructor-Led / Self-Paced**

# Course syllabus

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## COURSE OVERVIEW

This three-day course teaches students how to perform Transact-SQL queries, query analysis and query optimization on SQL Server 2025. Students will learn to use SQL Server Management Studio 22 and explore modern features including native JSON, vector search, and Intelligent Query Processing. Many lessons also apply to Azure SQL Database environments.

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## AT COURSE COMPLETION

After completing this course, you will be able to:

Write and execute Transact-SQL queries against SQL Server 2025 (version 17) databases using SSMS 22 and Copilot assistance

Analyze query performance using execution plans, SET STATISTICS IO, and AI-powered interpretation tools

Optimize queries through indexing strategies including clustered, nonclustered, covering, and JSON indexes

Leverage SQL Server 2025 (version 17) features including native JSON, Intelligent Query Processing, and Optimized Locking

Apply analytical query techniques using window functions, subqueries, and table expressions for complex reporting

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## PREREQUISITES

- Basic understanding of relational database concepts
  - Familiarity with Windows operating system
  - Basic computer navigation skills
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## COURSE OUTLINE

### MODULE 1 Querying Data with SQL Server Management Studio 22

115 min

Explore SQL Server Management Studio 22 and establish connections to SQL Server 2025 (version 17) instances. Learn to navigate the interface, execute queries, and leverage GitHub Copilot for query assistance.

#### LESSONS

- Lesson 1:** Introduction to SQL Server Management Studio 22
- Lesson 2:** Installing and Launching SSMS 22
- Lesson 3:** Connecting to SQL Server Instances
- Lesson 4:** Navigating Object Explorer
- Lesson 5:** Query Window Features and Result Management
- Lesson 6:** GitHub Copilot Integration for SQL Development
- Lesson 7:** Saving and Organizing Query Scripts

#### LAB EXERCISES

- Connecting to SQL Server 2025 and Executing Your First Queries

#### AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Connect to SQL Server 2025 (version 17) instances using SSMS 22
- Navigate Object Explorer and query execution windows
- Execute Transact-SQL queries and view results
- Use GitHub Copilot for query assistance and code generation

Master fundamental SELECT statements and explore SQL Server 2025 string manipulation functions and enhancements for text processing.

**LESSONS**

**Lesson 1:** Introduction to Transact-SQL and the SELECT Statement

**Lesson 2:** Column Aliasing and Expressions

**Lesson 3:** Essential String Functions: CONCAT, SUBSTRING, LEFT, and RIGHT

**Lesson 4:** String Manipulation Functions: TRIM, REPLACE, UPPER, LOWER, and REVERSE

**Lesson 5:** SQL Server 2025 String Enhancements

**Lesson 6:** NULL Handling in String Operations

**LAB EXERCISES**

String Manipulation and Data Transformation in SQL Server 2025

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**AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO**

Write SELECT statements with column selection and aliasing

Apply string manipulation functions including CONCAT, SUBSTRING, and TRIM

Utilize SQL Server 2025 (version 17) string enhancements

Format and transform string data for reporting

Learn to refine query results using WHERE clauses and ORDER BY to filter and sort data effectively.

**LESSONS**

**Lesson 1:** Introduction to Data Filtering and Sorting

**Lesson 2:** WHERE Clause Fundamentals and Comparison Operators

**Lesson 3:** Logical Operators:  
Combining Multiple  
Conditions

**Lesson 4:** Pattern Matching with LIKE  
and Wildcards

**Lesson 5:** Range and List Filtering with  
BETWEEN and IN

**Lesson 6:** Handling NULL Values with IS  
NULL and IS NOT NULL

**Lesson 7:** Sorting Results with ORDER  
BY

**LAB EXERCISES**

Filtering and Sorting Customer Order  
Data

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**AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO**

Implement WHERE clauses with comparison and logical operators

Filter data using LIKE, IN, BETWEEN, and IS NULL predicates

Sort query results using ORDER BY with multiple columns

Combine filtering and sorting for precise data retrieval

Master JOIN operations to combine data from multiple tables using INNER, OUTER, CROSS, and self-join techniques.

**LESSONS**

- Lesson 1:** Understanding Relational Relationships and JOIN Fundamentals
- Lesson 2:** INNER JOIN: Retrieving Matching Rows
- Lesson 3:** LEFT and RIGHT OUTER JOINS: Including Unmatched Rows
- Lesson 4:** FULL OUTER JOIN and CROSS JOIN Operations
- Lesson 5:** Self-Joins for Hierarchical and Recursive Data
- Lesson 6:** Multi-Table Joins and Join Order Optimization

**LAB EXERCISES**

- Combining Sales Data with Multiple JOIN Operations

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**AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO**

- Construct INNER JOIN queries to combine related tables
- Implement LEFT, RIGHT, and FULL OUTER JOIN operations
- Use CROSS JOIN for Cartesian products
- Apply self-join techniques for hierarchical data

## MODULE 5 Subqueries and Table Expressions

115 min

Explore subqueries, Common Table Expressions (CTEs), and derived tables for complex data retrieval and transformation.

### LESSONS

- Lesson 1:** Understanding Subqueries and Table Expressions
- Lesson 2:** Scalar and Multi-Row Subqueries
- Lesson 3:** Correlated Subqueries and Performance Considerations
- Lesson 4:** Common Table Expressions (CTEs)
- Lesson 5:** Recursive CTEs for Hierarchical Data
- Lesson 6:** Derived Tables and Table Expression Best Practices

### LAB EXERCISES

- Building Subqueries and Table Expressions

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### AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Write correlated and non-correlated subqueries
- Create and utilize Common Table Expressions (CTEs)
- Build derived tables and inline views
- Choose appropriate table expression techniques for specific scenarios

Master window functions including ranking, aggregate, and offset functions for advanced analytical queries.

**LESSONS**

**Lesson 1:** Introduction to Window Functions

**Lesson 2:** Ranking Functions: ROW\_NUMBER, RANK, and DENSE\_RANK

**Lesson 3:** Aggregate Window Functions and PARTITION BY

**Lesson 4:** Offset Functions: LAG, LEAD, FIRST\_VALUE, and LAST\_VALUE

**Lesson 5:** Window Frame Specifications: ROWS and RANGE

**Lesson 6:** Analytical Query Patterns for Business Intelligence

**LAB EXERCISES**

Analyzing Sales Data with Window Functions

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**AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO**

Apply ranking window functions including ROW\_NUMBER, RANK, and DENSE\_RANK

Use aggregate window functions with PARTITION BY and ORDER BY

Implement offset functions including LAG, LEAD, FIRST\_VALUE, and LAST\_VALUE

Construct complex analytical queries for business intelligence

## MODULE 7 Modern Data: Native JSON and Vector Search

138 min

Explore SQL Server 2025 (version 17) native JSON capabilities and vector search features for modern data workloads.

### LESSONS

- Lesson 1:** Introduction to Modern Data in SQL Server 2025
- Lesson 2:** Storing and Validating JSON Data
- Lesson 3:** Querying JSON with JSON\_VALUE and JSON\_QUERY
- Lesson 4:** Parsing JSON Arrays with OPENJSON
- Lesson 5:** Modifying and Generating JSON
- Lesson 6:** Introduction to Vector Search in SQL Server 2025

### LAB EXERCISES

- Working with Native JSON and Vector Search in SQL Server 2025

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### AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Query JSON data using native JSON functions in SQL Server 2025 (version 17)
- Create and modify JSON documents with JSON\_MODIFY and FOR JSON
- Store and retrieve JSON data in table columns
- Perform vector similarity searches using SQL Server 2025 (version 17) vector search capabilities

Learn to generate, interpret, and optimize query execution plans using SSMS 22 and AI-powered analysis tools.

**LESSONS**

**Lesson 1:** Understanding Execution Plans and Query Optimization

**Lesson 2:** Generating Execution Plans in SSMS 22

**Lesson 3:** Reading Execution Plans: Operators and Flow

**Lesson 4:** Scan vs. Seek: Understanding Data Access Patterns

**Lesson 5:** Performance Metrics: SET STATISTICS IO and TIME

**Lesson 6:** AI-Powered Execution Plan Analysis in SSMS 22

**LAB EXERCISES**

Analyzing Execution Plans with AI-Powered Tools

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**AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO**

Generate and interpret graphical execution plans in SSMS 22

Analyze query performance using SET STATISTICS IO and SET STATISTICS TIME

Use AI-powered tools to interpret execution plan recommendations

Identify expensive operators and resource consumption patterns

## MODULE 9 Indexing Strategies for Modern Workloads

134 min

Master index design and implementation including clustered, nonclustered, covering, and specialized indexes for SQL Server 2025 (version 17).

### LESSONS

- Lesson 1:** Understanding Database Indexes
- Lesson 2:** Clustered Index Fundamentals and Design
- Lesson 3:** Nonclustered Indexes and Covering Strategies
- Lesson 4:** Specialized Indexes: Filtered, Columnstore, and JSON
- Lesson 5:** Index Maintenance and Fragmentation Management
- Lesson 6:** Index Usage Analysis and Design Optimization

### LAB EXERCISES

- Implementing and Optimizing Index Strategies

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### AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Design and create clustered indexes for optimal data organization
- Implement nonclustered and covering indexes to improve query performance
- Create specialized indexes for JSON data and modern workload patterns
- Evaluate index usage and identify missing index opportunities

Explore Intelligent Query Processing features in SQL Server 2025 (version 17) including Optimized Locking and parameter sensitivity plan optimization.

**LESSONS**

- Lesson 1:** Introduction to Intelligent Query Processing
- Lesson 2:** Database Compatibility Level Requirements
- Lesson 3:** Optimized Locking in SQL Server 2025
- Lesson 4:** Parameter Sensitive Plan Optimization
- Lesson 5:** Cardinality Estimation and Memory Grant Feedback
- Lesson 6:** Adaptive Joins and Batch Mode Processing
- Lesson 7:** Monitoring IQP with Query Store
- Lesson 8:** IQP-Related DMVs and System Views
- Lesson 9:** IQP Best Practices
- Lesson 10:** Troubleshooting IQP Issues

**LAB EXERCISES**

- Implementing and Monitoring Intelligent Query Processing Features

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**AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO**

- Enable and configure Intelligent Query Processing features in SQL Server 2025 (version 17)
- Leverage Optimized Locking to reduce blocking and improve concurrency
- Apply parameter sensitivity plan optimization for variable workloads
- Monitor IQP feature activation using Query Store and system views

## BOOTCAMP MODE — CAPSTONE LABS

One integrative Capstone Lab is generated per course day. Each capstone presents a new scenario requiring students to independently apply the combined skills from that day's modules — no procedure, no hints. Students determine their own approach. Capstone Labs are the afternoon activity in Bootcamp delivery and replace the lecture portion of standard Instructor-Led Training.

### DAY 1 CAPSTONE Modules 1, 2, 3

#### SKILLS TO BE INTEGRATED

- Connect to SQL Server 2025 (version 17) instances using SSMS 22 and execute queries
- Navigate Object Explorer and query windows to interact with database objects
- Use GitHub Copilot for query assistance and code suggestions
- Write SELECT statements with column selection and aliasing
- Apply string manipulation functions including CONCAT, SUBSTRING, and TRIM
- Utilize SQL Server 2025 string enhancements for text processing
- Implement WHERE clauses with comparison and logical operators
- Sort query results using ORDER BY with multiple columns and directions
- Filter data using LIKE patterns, IN lists, and BETWEEN ranges

### DAY 2 CAPSTONE Modules 4, 5, 6, 7

#### SKILLS TO BE INTEGRATED

- Construct INNER JOIN queries to combine data from multiple tables
- Implement LEFT, RIGHT, and FULL OUTER JOIN operations
- Use CROSS JOIN and self-join techniques for complex relationships
- Write correlated and non-correlated subqueries in SELECT, WHERE, and FROM clauses
- Create and utilize Common Table Expressions (CTEs) for hierarchical data
- Build derived tables and inline views for data transformation
- Apply window functions including ROW\_NUMBER, RANK, and DENSE\_RANK
- Use aggregate window functions with PARTITION BY and ORDER BY
- Implement LAG, LEAD, and other offset functions for analytical queries
- Query JSON data using native JSON functions in SQL Server 2025 (version 17)
- Create and modify JSON documents with JSON\_MODIFY and FOR JSON
- Perform vector similarity searches using SQL Server 2025 vector search capabilities

### **DAY 3 CAPSTONE** Modules 8, 9, 10

#### **SKILLS TO BE INTEGRATED**

- Generate and interpret graphical execution plans in SSMS 22
- Analyze query performance metrics using SET STATISTICS IO and TIME
- Use AI-powered tools to interpret execution plan bottlenecks and recommendations
- Identify expensive operators and resource consumption patterns
- Design and create clustered indexes for optimal data retrieval
- Implement nonclustered and covering indexes to improve query performance
- Create specialized indexes for JSON data and modern workload patterns
- Evaluate index usage statistics and identify missing indexes
- Enable and configure Intelligent Query Processing features in SQL Server 2025 (version 17)
- Leverage Optimized Locking to reduce blocking and improve concurrency
- Apply parameter sensitivity plan optimization for variable workloads
- Monitor IQP feature activation using system views and Query Store

*Capstone content is embedded in the student manuals and instructor guides — one section per day, after the final module for that day. A standalone Capstone Slides (.pptx) file is also included in the Exclusive download package.*